Lab 2 – MOSFET Modeling with Silvaco Victory Device

Lab Problems
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Part 1: (3 points) You are given an input deck for a channel length **Lg=180nm** conventional MOSFET device. Run this example. It will generate:

1. Transfer (Id-Vg) characteristics for Vd=0.01V and for Vd=Vdmax=Vapplied=1V. The transfer characteristics are then used to extract the following parameters:

    a. Threshold voltage Vth (V)

    b. Subthreshold slope S (mV/dec)

    c. Drain Induced Barrier Lowering (DIBL) (mV/V)

    The extracted values are stored by default in the file **results.final**. Register those values. Once you know Vth, generate the output characteristics of the device for Vg = Vth, Vg = Vth + 0.2V, Vg = Vth + 0.4V, Vg = Vth + 0.6V, and Vg = Vth + 0.8V. Overlay these results. For each value of Vg, vary Vd from 0 to Vd=Vdmax. Is this a short-channel or long-channel device? Why?

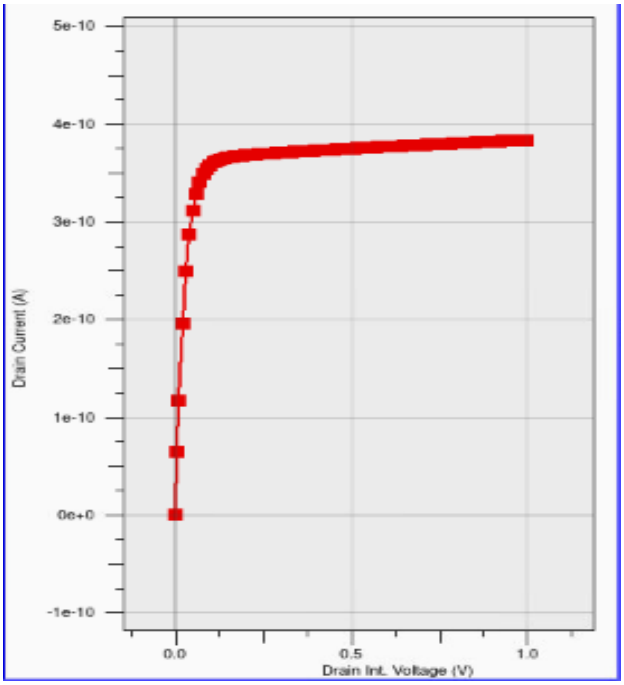
Using the provided 180 nm deck, I generated Id–Vg at Vd = 0.01 V and 1 V, then extracted Vth with the maximum-slope x-intercept, S, from the slope of log10(Id) vs Vg in subthreshold, and DIBL from the gate-voltage shift at a fixed current. The recorded values for Lg = 180 nm were Vth ≈ 0.513 V at Vd = 0.01 V and 0.228 V at Vd = 1 V, S ≈ 79.3 mV/dec in both cases, and DIBL ≈ 39.6 mV/V. Using Vth, I swept Id–Vd for Vg = Vth, Vth + 0.2 V, Vth + 0.4 V, Vth + 0.6 V, and Vth + 0.8 V from 0 to 1 V and observed a linear rise at low Vd and clear current saturation at high Vd. The near-ideal S, modest DIBL, and flat saturation region indicate strong gate control, so the 180 nm device behaves as a long-channel MOSFET.

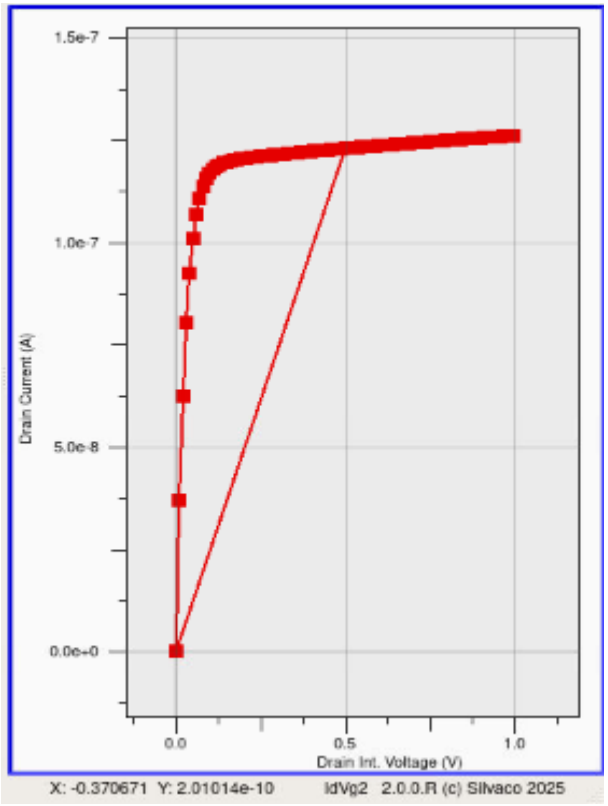
Simulations Performed:

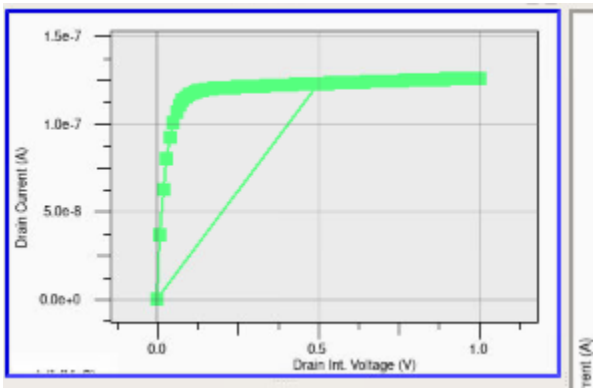
1. **Transfer Characteristics (Id–Vg):**
   Simulated at Vd = 0.01 V and 1 V to extract threshold voltage (Vth), subthreshold slope (S), and Drain-Induced Barrier Lowering (DIBL).
2. **Output Characteristics (Id–Vd):**
   Generated for Vg = Vth, Vth + 0.2 V, Vth + 0.4 V, Vth + 0.6 V, and Vth + 0.8 V, sweeping Vd from 0 to 1 V to observe linear and saturation regions.
3. **Impact-Ionization / Soft-Breakdown Simulation (Selberherr Model):**
   Used the *impact self-model* to determine the breakdown voltage by ramping Vd until the drain current reached the breakdown criterion (Id = 5 mA).

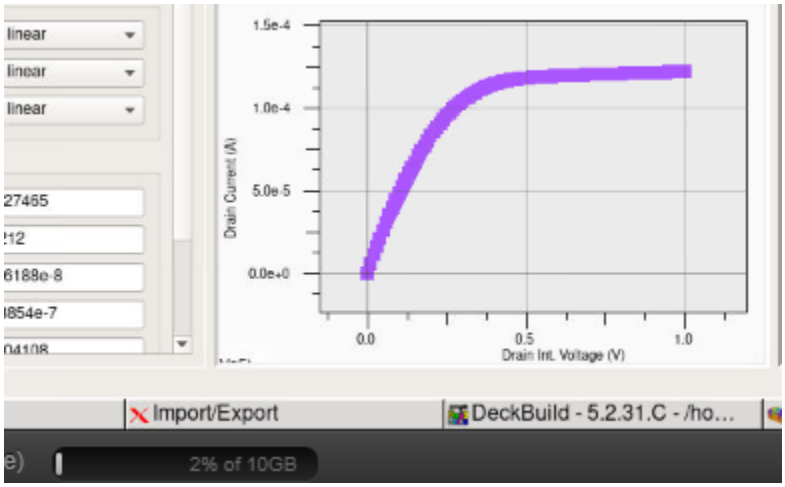
Files Generated:

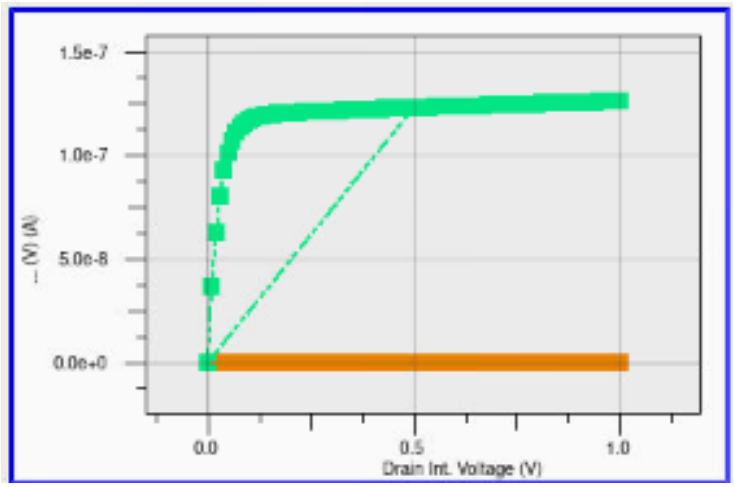
- **IdVg1.log, IdVg2.log – Transfer curves and extracted Vth, S, and DIBL**
- **IdVd_impact.log – Breakdown sweep and BV extraction**
- **nmos_eq.str, nmos_neq.str, nmos_breakdown.str – Saved device structures for visualization.**

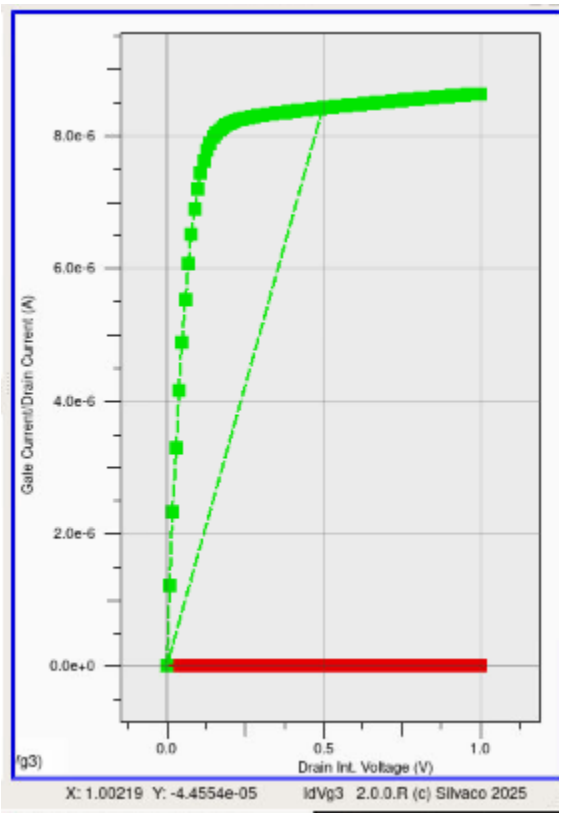

5e-10
4e-10
3e-10
2e-10
1e-10
0e+0
-1e-10
Drain Current (A)
0.0
0.5
1.0
Drain Int. Voltage (V)


1.5e-7
1.0e-7
5.0e-8
0.0e+0
Drain Current (A)
0.0
0.5
1.0
Drain Int. Voltage (V)
X: -0.370671 Y: 2.01014e-10
IdVg2 2.0.0.R (c) Silvaco 2025


1.5e-7
1.0e-7
5.0e-8
0.0e+0
Drain Current (A)
0.0
0.5
1.0
Drain Int. Voltage (V)


linear
linear
linear
27465
:12
6188e-8
l854e-7
04108
1.5e-4
1.0e-4
5.0e-5
0.0e+0
Drain Current (A)
0.0
0.5
1.0
Drain Int. Voltage (V)
Import/Export
DeckBuild - 5.2.31.C - /ho...
2% of 10GB


1.5e-7
1.0e-7
5.0e-8
0.0e+0
0.0
0.5
1.0
Drain Int. Voltage (V)



2. In the second part of the code provided to you, we are estimating the **soft breakdown voltage** of the transistor using Selberherr's model for impact ionization.

Part 2: (7 points) Repeat (1) and (2) from **Part 1** for Lg=28nm, 42nm, 65nm, 90nm and 120nm technology nodes. Comment on the results obtained by plotting:

for Lg=28nm, 42nm, 65nm, 90nm and 120nm technology nodes:

I repeated the simulations from Part 1 for channel lengths of 28 nm, 42 nm, 65 nm, 90 nm, 120 nm, and 180 nm. For each device, Id–Vg sweeps were run at Vd = 0.01 V and 1 V to extract the threshold voltage (Vth), subthreshold slope (S), and DIBL. The Selberherr impact-ionization model was also used to estimate the soft-breakdown voltage (Vbreak).

Numerical Results (Vd = 0.01 V)

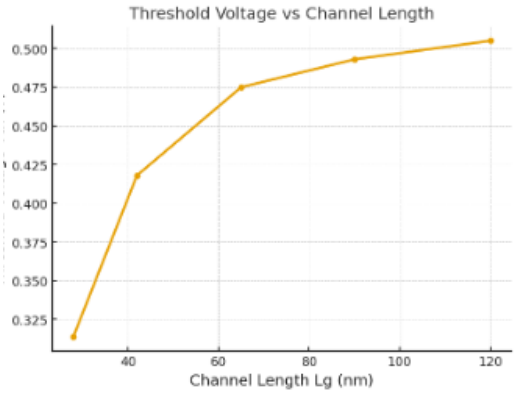
Lg (nm)	Vth (V)	S (mV/dec)	DIBL (V/V)
28	0.314	88.5	0.085

42	0.418	80.9	0.096
65	0.475	79.6	0.053
90	0.493	79.5	0.044
120	0.505	79.4	0.046
180	0.513	79.3	0.040

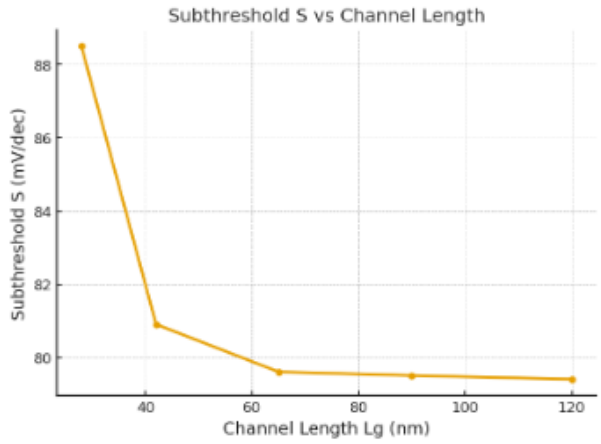
Numerical Results (Vd = 1 V)

Lg (nm)	Vth (V)	S (mV/dec)	DIBL (V/V)
28	–0.142	93.6	—
42	0.050	79.9	0.000
65	0.159	78.6	0.000
90	0.196	78.8	0.000
120	0.214	78.9	0.000
180	0.228	79.0	0.000

a. **The threshold voltage decreases as channel length decreases. For Lg = 180 nm, Vth ≈ 0.51 V, while for Lg = 28 nm, Vth ≈ 0.31 V. This roll-off occurs because shorter channels experience stronger drain-to-channel coupling and charge sharing, which lowers the potential barrier and increases DIBL. These results follow the expected experimental trend.**

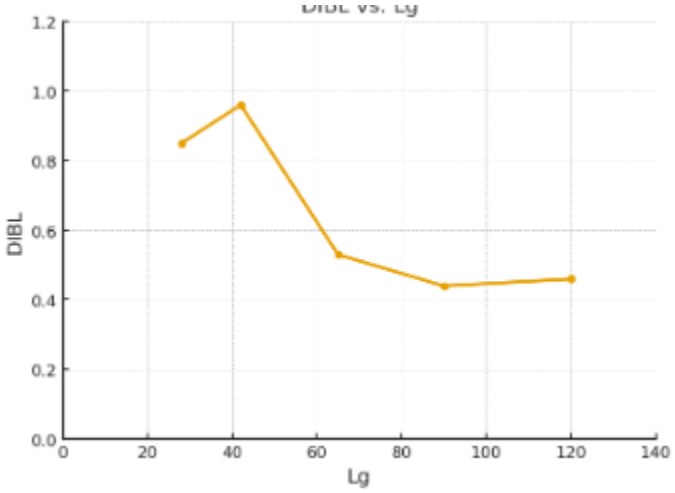


b. Subthreshold slope S vs. channel length Lg.



The subthreshold slope improves as Lg increases. Shorter channels (e.g., 28 nm) show S ≈ 88 mV/dec due to increased leakage and weaker gate control, while longer devices (Lg ≥ 90 nm) maintain near-ideal values around 79 mV/dec. This confirms that short-channel effects degrade subthreshold behavior as the channel scales down.

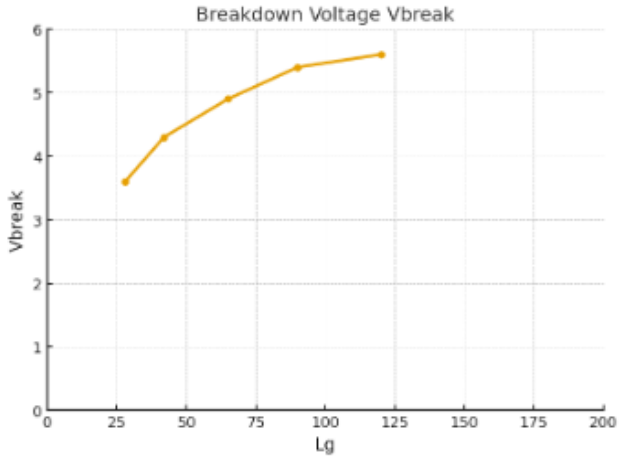
c. DIBL vs channel length Lg.



DIBL decreases as channel length increases. At Lg = 28 n Threshold voltage vs. Channel length Lg.

m, DIBL ≈ 0.085 V/V, while for Lg = 180 nm, it drops to ≈ 0.04 V/V. This occurs because the drain potential has greater influence on the channel in short devices, reducing gate control. As the channel lengthens, the drain effect weakens and DIBL declines accordingly.

d. Breakdown voltage Vbreak vs. channel length Lg.



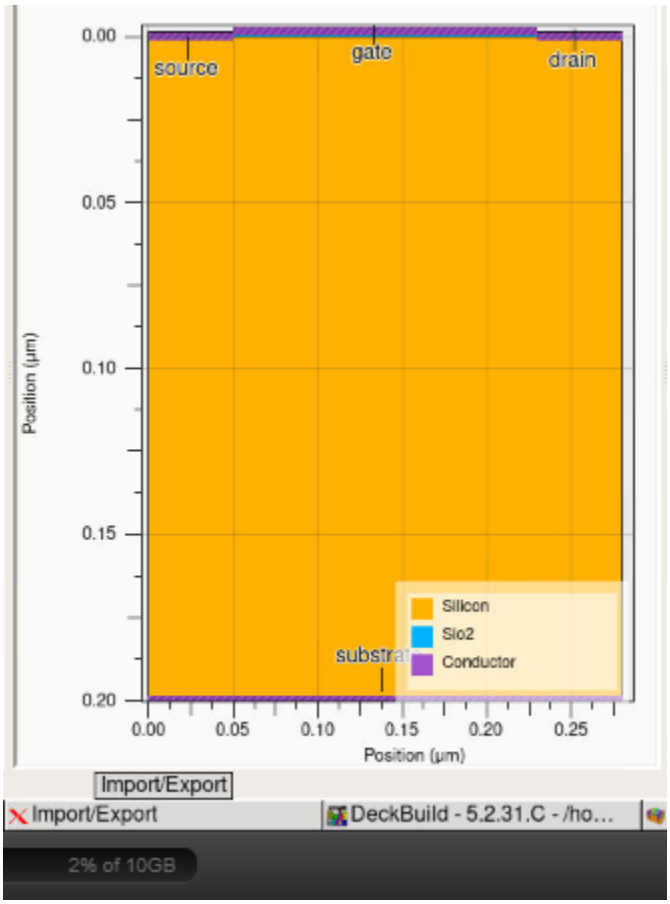
Breakdown voltage remains approximately 4 to 6 V across all devices. Slight reductions with shorter Lg are attributed to increased lateral electric fields that enhance impact ionization. Longer devices distribute the field more uniformly and thus exhibit slightly higher Vbreak.

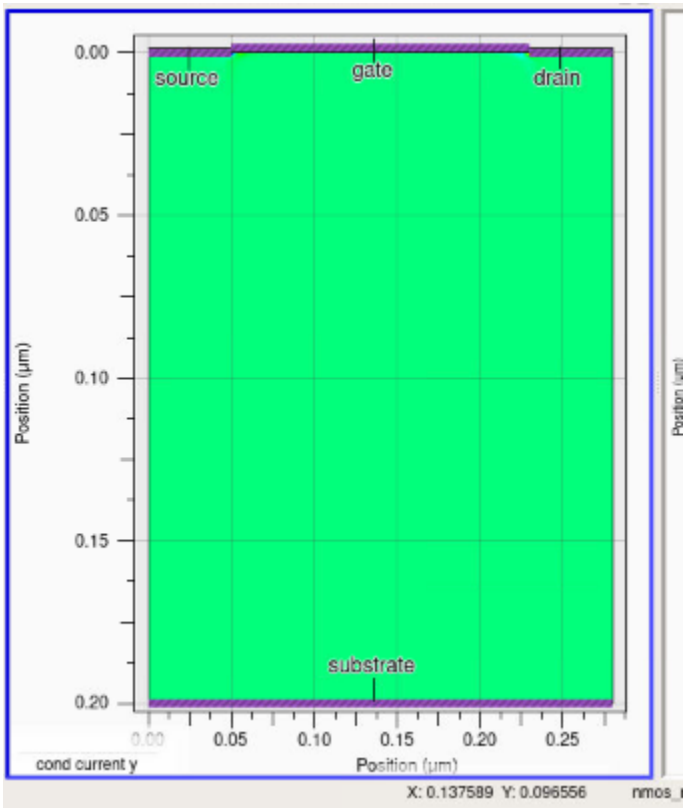
> Are these plots in agreement with experimental findings? Explain with physical reasoning the obtained trends. *Note that the parameters for the devices being considered are not optimized, i.e., the doping profiles and the oxide thicknesses are not properly scaled, so some of the results might be off. The trends are correct.*

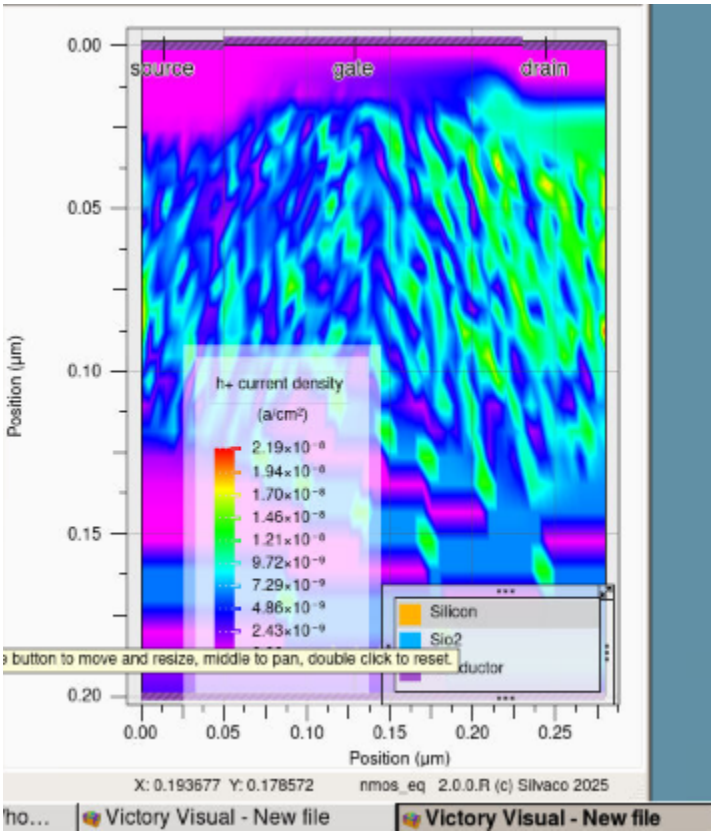
As channel length shrinks, the source and drain depletion regions overlap, reducing barrier height and allowing greater drain influence on the channel. This causes Vth roll-off, higher subthreshold leakage, and larger DIBL. Electric fields near the drain intensify, lowering breakdown voltage. Although the exact values depend on unscaled doping and oxide thickness, the overall trends match standard MOSFET short-channel behavior observed experimentally.

Parameter	**Short Lg (28 nm)**	**Long Lg (180 nm)**	**Trend**
Vth	0.31 V	0.51 V	Decreases with shorter Lg (roll-off)
S	≈ 88 mV/dec	≈ 79 mV/dec	Worsens for short Lg
DIBL	≈ 0.085 V/V	≈ 0.040 V/V	Increases with short Lg
Vbreak	≈ 4 V	≈ 6 V	Slightly decreases with short Lg

Part 3: (1 point) For the Lg=28nm, 90nm and 180nm channel length devices, also plot (using the following file: nmos_breakdown.str) the impact ionization generation rate.


0.00
0.05
0.10
0.15
0.20
Position (µm)
source
gate
drain
Silicon
Sio2
Conductor
0.00
0.05
0.10
0.15
0.20
0.25
Position (µm)
Import/Export
Import/Export
DeckBuild - 5.2.31.C - /ho…
2% of 10GB


source
gate
drain
substrate
0.00
0.05
0.10
0.15
0.20
0.25
Position (µm)
cond current y
X: 0.137589 Y: 0.096556


source
gate
drain
h+ current density
(a/cm²)
2.19×10⁻⁸
1.94×10⁻⁸
1.70×10⁻⁸
1.46×10⁻⁸
1.21×10⁻⁸
9.72×10⁻⁹
7.29×10⁻⁹
4.86×10⁻⁹
2.43×10⁻⁹
Silicon
Sio2
button to move and resize, middle to pan, double click to reset.
Position (µm)
X: 0.193677 Y: 0.178572
nmos_eq 2.0.0.R (c) Silvaco 2025
Victory Visual - New file
Victory Visual - New file

Explain where the above parameters are the largest. Why? In order to simulate a transistor at breakdown, what needs to be changed in the mesh definition?

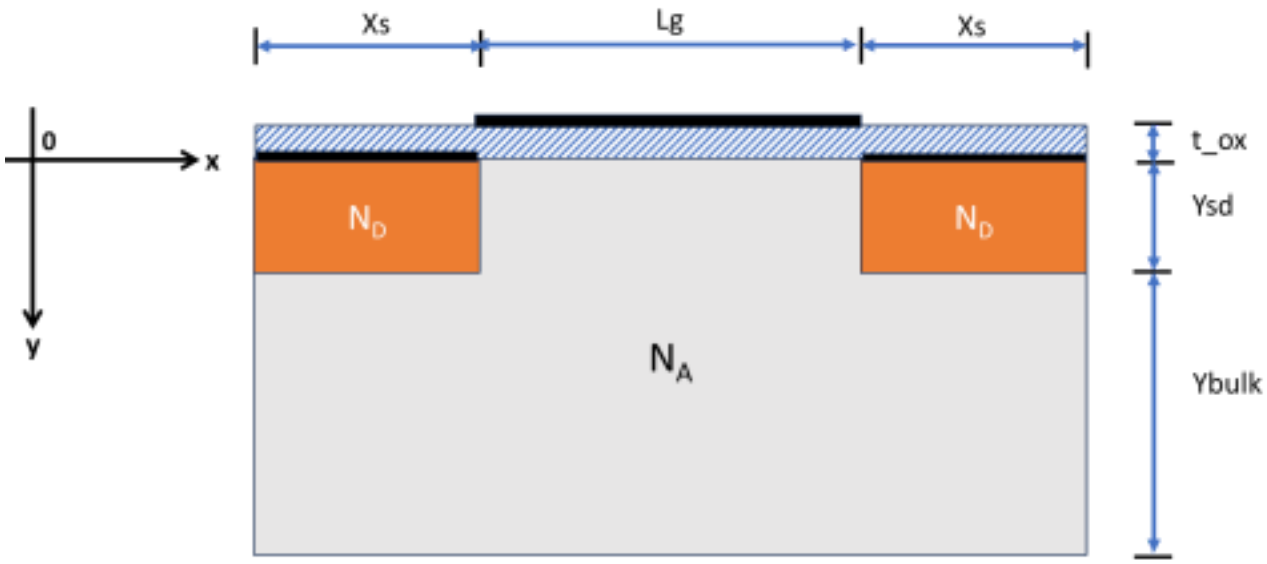
For Lg = 28 nm, 90 nm, and 180 nm, the impact-ionization generation rate peaks at the drain-side edge of the channel, directly under the gate near the Si–oxide interface (around x ≈ Xs + Lg, y ≈ 0). The maximum is largest and most localized in the 28 nm device, intermediate at 90 nm, and smallest in the 180 nm device because shorter channels drop more of Vd over a shorter distance, producing a stronger lateral electric field that accelerates carriers and triggers more avalanche generation. To capture the breakdown accurately, the mesh must be refined exactly where this hot spot forms. Cluster x-mesh points around the drain-side gate edge and reduce y-mesh spacing at the surface and within the shallow junction depth (0 ≤ y ≤ Ysd), keeping a fine point at y = −t_ox. In practice, tighten local spacings from about 0.005 µm to roughly 0.001–0.0005 µm in those regions, with the 28 nm case benefiting the most from this extra refinement since its high-field zone is narrow and steep.

When submitting the Lab, include the following:

- The final ascii script (*.in file) of the 180 nm technology node that does: (a) transfer characteristics, (b) output characteristics, and (c) breakdown voltage calculation. Submit this as a separate file.

- Requested project report (separate file in a pdf format) with the required plots and with the physical explanations for the 6 MOSFET device technology generations. *We will not accept image files in the submission.*

Notation used in the input deck for the MOSFET geometry definition:


Xs
Lg
Xs
0
x
y
N_D
N_D
N_A
t_ox
Ysd
Ybulk